

Trial Examination 2025

HSC Year 12 Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A data sheet, formulae sheet and Periodic Table are provided at the back of this paper

Total Marks: 100

Section I – 20 marks (pages 2–10)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 11–27)

- Attempt Questions 21–35
- Allow about 2 hours and 25 minutes for this section

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2025 HSC Year 12 Physics examination.

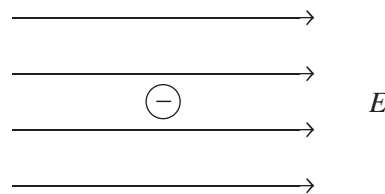
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SECTION I**20 marks****Attempt Questions 1–20****Allow about 35 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–20.

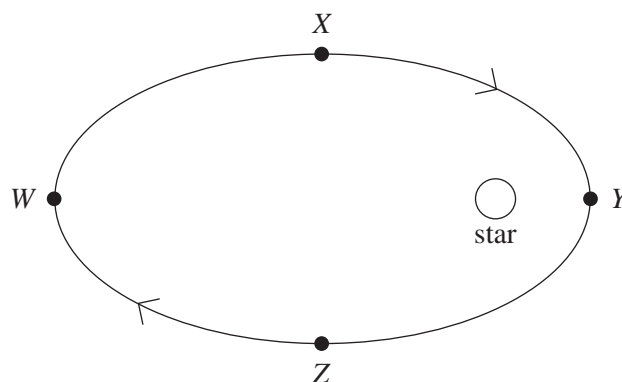
- 1 Which of the following would decrease the time of flight of a ball that falls to the ground after rolling off a horizontal bench?
- A. increasing the speed of the ball
 - B. decreasing the speed of the ball
 - C. increasing the height of the bench
 - D. decreasing the height of the bench

- 2 A negatively charged particle is stationary in an electric field, E , as shown in the diagram.



What is the direction of the force acting on the particle due to the electric field?

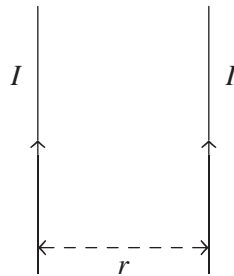
- A. left
 - B. right
 - C. up
 - D. down
- 3 A comet is in a stable elliptical orbit around a star, as shown in the diagram.



At which position does the comet have the greatest speed?

- A. W
- B. X
- C. Y
- D. Z

- 4 What is the peak wavelength of the radiation emitted by a black body with a temperature of 3500 K?
- A. 5.47×10^{-7} m
B. 8.28×10^{-7} m
C. 2.86×10^{-4} m
D. 1.21×10^6 m
- 5 Which of the following is a quark composition hadron?
- A. muon
B. proton
C. lepton
D. photon
- 6 Two infinitely long parallel wires, each carrying the same current, I , are separated by a distance, r , as shown in the diagram. This produces a force, F , on each wire.

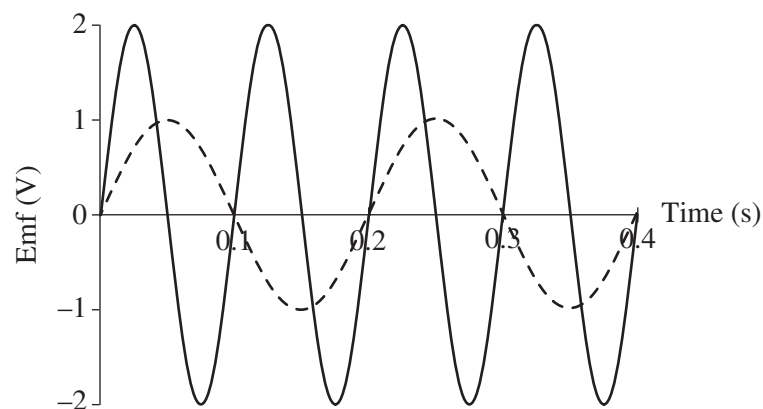


The distance between the wires is increased to $2r$.

What is the magnitude of the force on each wire after this change?

- A. $\frac{F}{4}$
B. $\frac{F}{2}$
C. F
D. $2F$
- 7 Two radioactive samples, W and X , initially have an activity ratio $\frac{W_{\text{activity}}}{X_{\text{activity}}}$ of 3.
- If each sample has the same half-life, what is the activity ratio after THREE half-lives?
- A. 1
B. 3
C. 6
D. 9

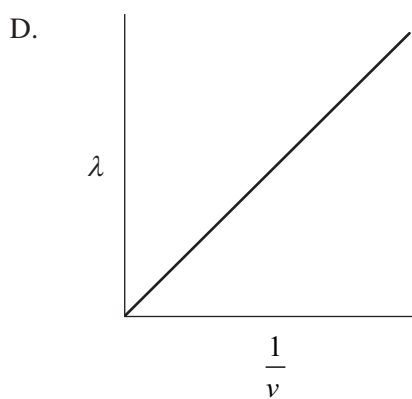
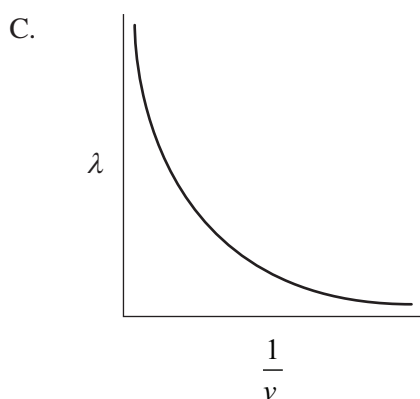
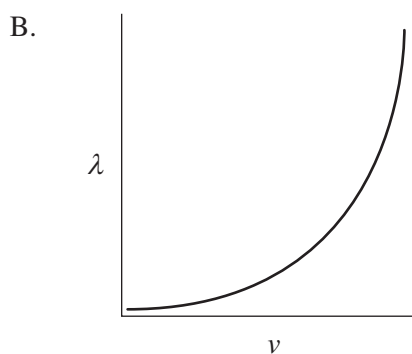
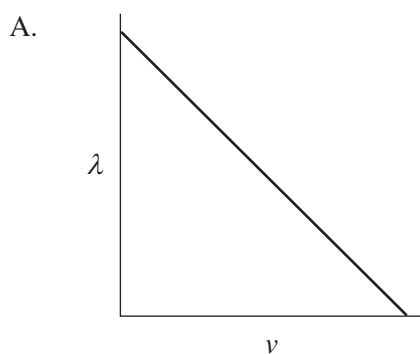
- 8 A commutator ensures continuous rotational motion in a DC motor by
- A. reducing friction between the rotor and the stator.
 - B. reversing the current direction in the rotor every half-turn.
 - C. maintaining a constant voltage across the rotor at all times.
 - D. providing a fixed magnetic field to the rotor throughout its rotation.
- 9 A simple AC generator produces an output emf according to the solid line on the graph shown.



Which modification to the generator would result in the output shown by the dashed line?

- A. halving the area of the coil
- B. halving the number of turns in the coil
- C. halving the speed of rotation of the coil
- D. halving the strength of the external magnetic field

10 Which graph represents the relationship between the de Broglie wavelength, λ , and speed, v ?



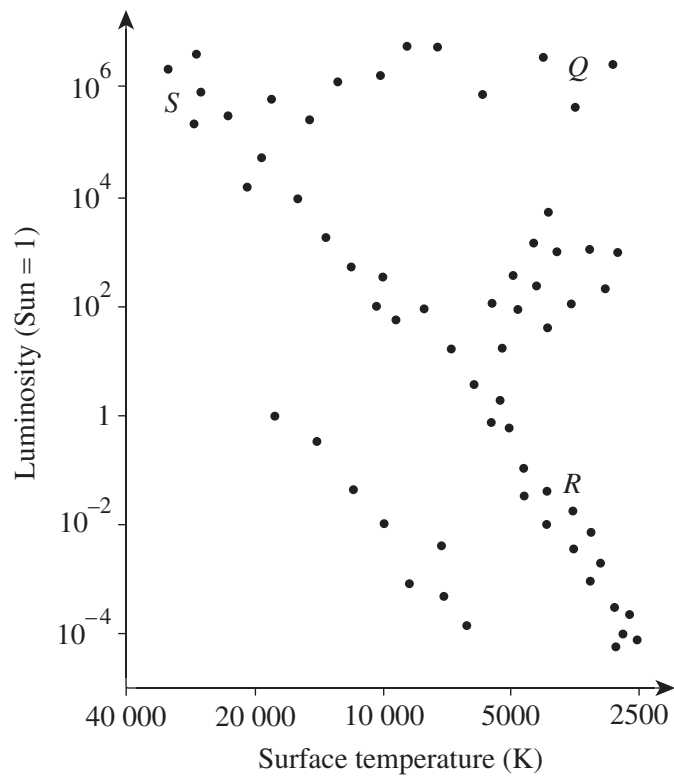
11 An electron and an antineutrino are emitted from a plutonium-241 nucleus during radioactive decay. Which of the following is the daughter nucleus that results from this decay?

- A. neptunium-240
- B. americium-240
- C. neptunium-241
- D. americium-241

12 Which of the following would occur if the coils of an operating simple DC motor were suddenly stopped by an external force?

- A. increase in current
- B. decrease in net emf
- C. increase in back emf
- D. decrease in back current

- 13 A metal ball with a diameter of 12 cm rotates at a frequency of 3.5 Hz. What is the maximum speed that any point on the surface of the ball can obtain?
- A. 1.3 m s^{-1}
B. 1.7 m s^{-1}
C. 2.6 m s^{-1}
D. 3.4 m s^{-1}
- 14 A Hertzsprung–Russell diagram is shown. Three regions, Q , R and S , are labelled.

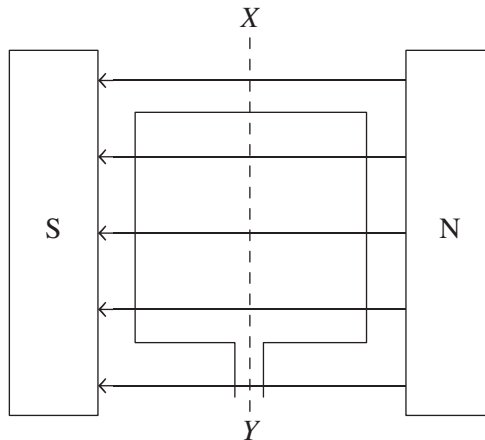


Which row of the table identifies the correct regions?

	<i>Region where the p-p chain is dominant in cores</i>	<i>Region where stars have the greatest size</i>
A.	R	Q
B.	R	S
C.	S	Q
D.	S	S

- 15 What is the energy of a gamma ray photon produced from a single positron–electron annihilation?
- A. 0.5 eV
 - B. 1.0 eV
 - C. 0.5 MeV
 - D. 1.0 MeV

- 16 A coil rotates around the X – Y axis at a constant speed in a uniform magnetic field, as shown in the diagram.

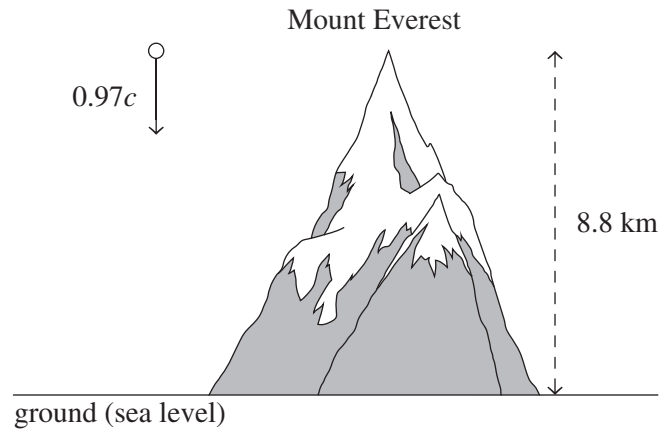


At the instant shown, the plane of the coil and the magnetic field are parallel. This means that the angle, θ , between them is 0° .

The magnitude of the induced emf in the coil is at a minimum

- A. when θ is 0° .
 - B. when θ is 45° .
 - C. when θ is 90° .
 - D. throughout the coil's rotation.
- 17 Two planets, A and B , orbit the same star. Planet A 's average orbital distance is six times larger than the average orbital distance of planet B .
- What is the relationship between the orbital periods, T , of the planets?
- A. $T_A = 6T_B$
 - B. $T_A = 15T_B$
 - C. $T_A = 36T_B$
 - D. $T_A = 216T_B$

- 18** At rest, the peak of Mount Everest has a height of 8.8 km above sea level. A cosmic ray collides with the atmosphere at the same altitude as the peak of Mount Everest, producing a muon. An observer at rest at the peak of Mount Everest measures the muon as travelling at a constant speed of $0.97c$ directly down towards the ground, which is at sea level, as shown in the diagram.

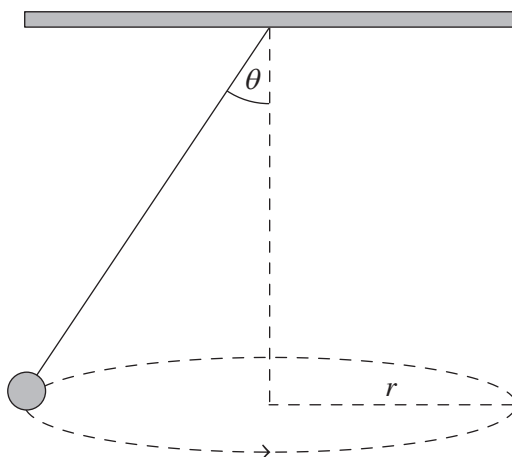


Muons last for $2.2 \mu\text{s}$ at rest before they decay.

From the muon's perspective, what approximate height above sea level would the muon reach before it decays?

- A. 0.6 km
- B. 1.5 km
- C. 4.4 km
- D. 8.2 km

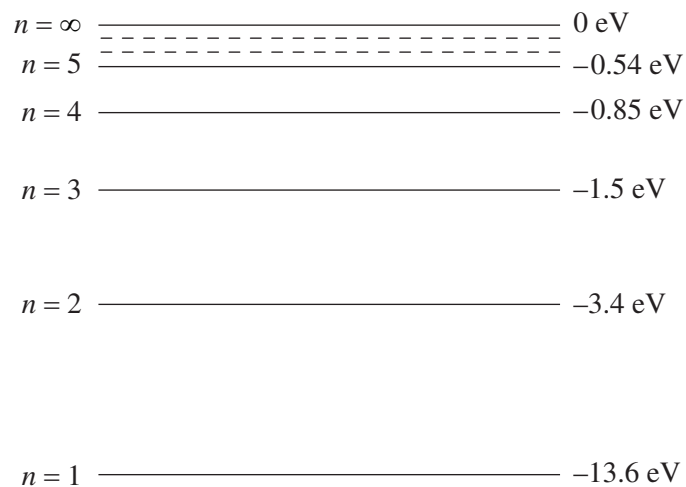
- 19 A ball that is attached to a ceiling with a thin string is undergoing uniform circular motion, as shown in the diagram.



Which of the following is the correct expression for calculating θ ?

- A. $\tan^{-1}\left(\frac{v^2}{rg}\right)$
- B. $\sin^{-1}\left(\frac{mg}{T}\right)$
- C. $\sin^{-1}\left(\frac{m^2v^2g}{r}\right)$
- D. $\tan^{-1}\left(\frac{mv^2}{rg}\right)$

20 The electron energy level diagram for a hydrogen atom is shown.



Which of the following statements is correct?

- A. An electron that transitions from $n = 2$ to $n = 5$ produces a photon with 2.86 eV of energy.
- B. The Balmer series is the name given to spectral emission lines that result from electron transitions from higher levels down to $n = 1$.
- C. The energy required to ionise a hydrogen atom when its electron is in the ground state is four times greater than the energy required to transition an electron from $n = 1$ to $n = 2$.
- D. An electron that transitions from $n = 4$ to $n = 2$ produces a photon with a wavelength that is longer than the wavelength of a photon produced from an electron that transitions from $n = 2$ to $n = 1$.

HSC Year 12 Physics

Section II Answer Booklet

80 marks

Attempt Questions 21–35

Allow about 2 hours and 25 minutes for this section

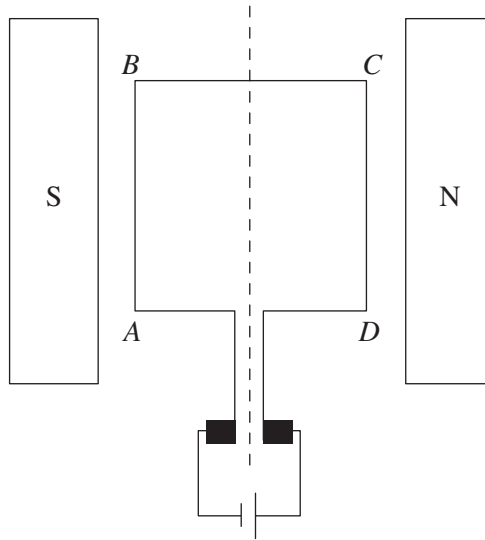
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Please turn over

Question 21 (4 marks)

The diagram shows a simple DC motor with a single loop of wire, $ABCD$.



- (a) Compare the forces acting on sides AB and CD in the position shown.

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- (b) Outline the effect that doubling the number of loops would have on the maximum torque produced by the motor. Support your answer with a relevant equation.

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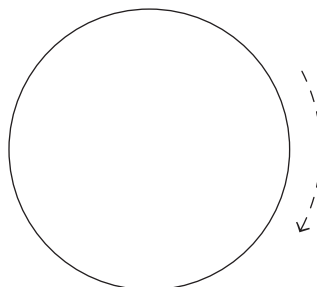
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Question 22 (5 marks)

A horizontal disc with a radius of 0.35 m rotates around its centre at a constant rate of one revolution per second, as shown in the diagram.



A 0.031 kg object is fixed to the edge of the disc.

- (a) Calculate the centripetal force acting on the object.

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- (b) What would happen to the angular speed of the object if it was moved to a point closer to the centre of the disc? Support your answer with a relevant equation.

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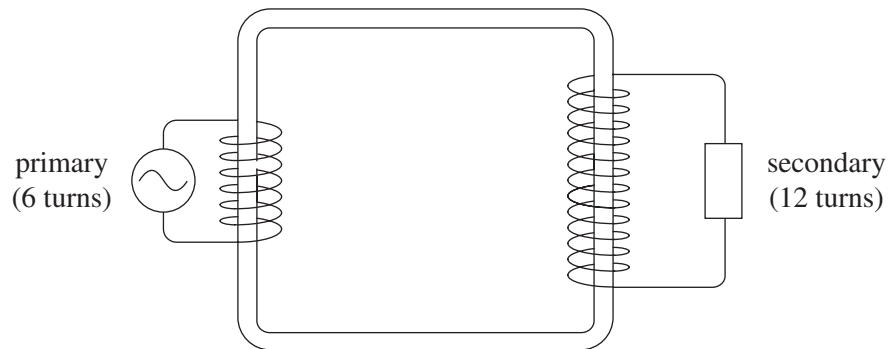
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Question 23 (6 marks)

The diagram shows an ideal transformer.



The primary coil is connected to an AC supply. The current in the secondary coil is 5 A.

- (a) Show that the current in the primary coil is 10 A.

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- (b) Explain why it is important to use AC rather than DC when operating transformers.

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Question 24 (4 marks)

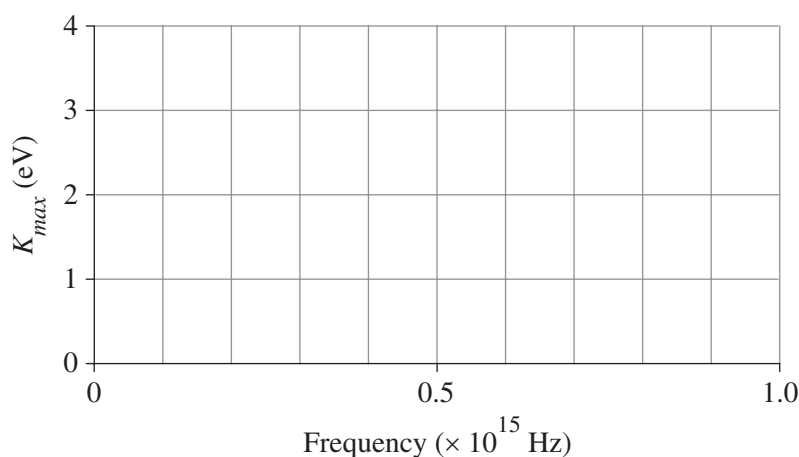
The surface of a metal was exposed to incident light of various frequencies. The maximum kinetic energy of the photoelectrons emitted from the metal at each frequency was measured.

4

The results of the investigation are recorded in the table.

<i>Frequency ($\times 10^{15}$ Hz)</i>	0.40	0.64	0.72	0.80	0.98
<i>K_{max} (eV)</i>	0.8	1.7	2.0	2.3	3.1

Plot the results on the axes provided and, hence, determine whether photoelectrons would be emitted from the metal if it was exposed to light with a wavelength of $3\text{ }\mu\text{m}$. Include a relevant calculation in your answer.



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Question 25 (3 marks)

Outline Schrödinger's contribution to the quantum mechanical model of the atom.

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Question 26 (6 marks)

- (a) In the Hafele–Keating experiment, two highly accurate atomic clocks were flown around the world on commercial flights – one flying eastward and one flying westward – and then compared to atomic clocks that remained on the ground in Washington D.C. The eastward flight moved in the direction of Earth’s rotation, while the westward flight moved against the direction of Earth’s rotation.

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Explain how the Hafele–Keating experiment validated time dilation as predicted by special relativity.

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- (b) The momentum of a particular electron is measured to be $8.7 \times 10^{-22} \text{ kg m s}^{-1}$. Explain why classical physics cannot accurately account for this momentum. Support your answer with a relevant calculation.

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Question 27 (5 marks)

With a radius of 476 km and mass of 9.1×10^{20} kg, the dwarf planet Ceres is the largest object in the asteroid belt between Mars and Jupiter.

- (a) If a 900 kg rover landed on the surface of Ceres, show that its gravitational potential energy would be -1.1×10^8 J. **2**

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- (b) Calculate the escape velocity from the surface of Ceres. **3**

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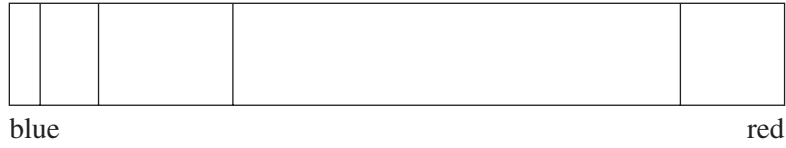
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Question 28 (7 marks)

The hydrogen absorption lines in a star's spectrum are shown in the diagram.



- (a) Account for the production of this spectrum.

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- (b) Explain how the translational velocity of this star could be determined from its spectrum.

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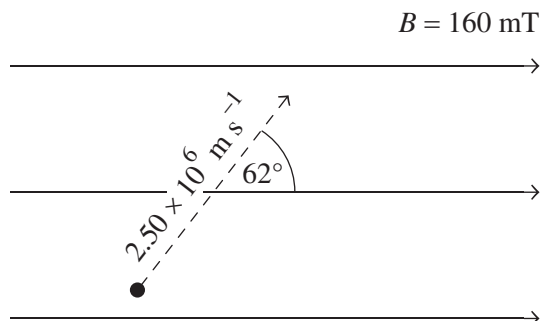
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Question 29 (4 marks)

A charged particle is moving at $2.50 \times 10^6 \text{ m s}^{-1}$ at an angle of 62.0° to the direction of a uniform magnetic field of strength 160 mT, as shown in the diagram.

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The particle moves in a helical path in the magnetic field with a radius of 0.286 m.

Calculate the charge-to-mass ratio of the particle.

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Question 30 (6 marks)

Evaluate the significance of polarisation in developing a model for light.

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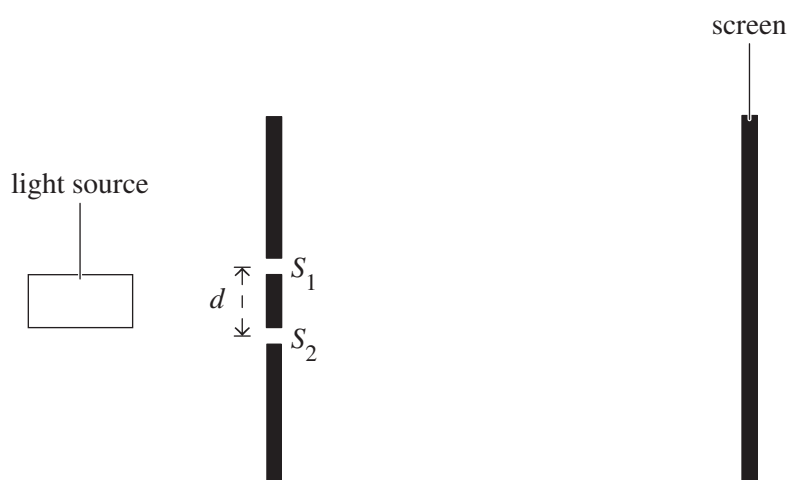
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Question 31 (4 marks)

The diagram illustrates an experiment where monochromatic light is incident upon a pair of thin slits, S_1 and S_2 , that are separated by a distance, d . A screen is placed behind the slits. **4**



A series of equidistant bright regions is observed on the screen.

When the experiment is repeated, the frequency of the incident light is increased.

Explain TWO modifications that can be made to the setup of this experiment to maintain the distances between the bright regions.

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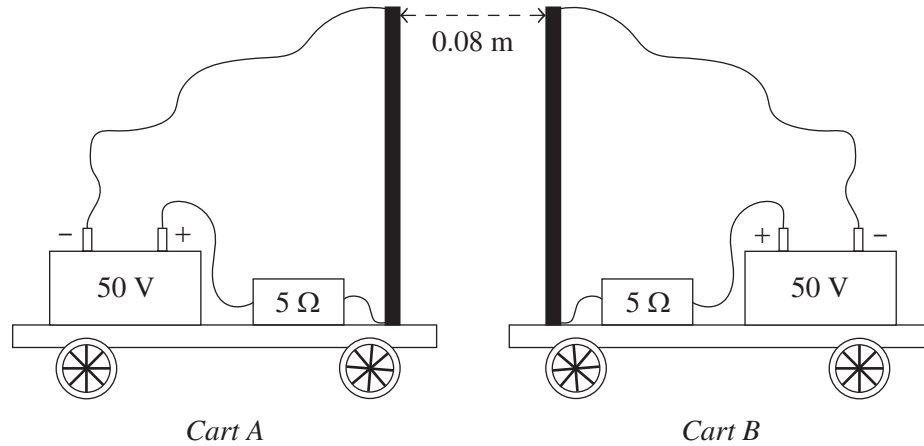
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Question 32 (5 marks)**5**

Two identical carts, *A* and *B*, of mass 0.65 kg each have a fixed 0.50 m long vertical metal rod connected to a 50 V battery and 5 Ω resistor.

Initially, the batteries are switched off, and the metal rods are separated by a distance of 0.08 m, as shown in the diagram.



Determine the initial acceleration of each cart when the batteries are switched on.

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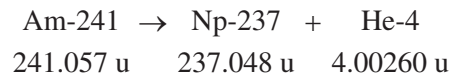
Analyse how the Geiger–Marsden experiment and Chadwick’s discovery of the neutron provide experimental evidence for the nuclear model of the atom.

8

[illegible]

Question 34 (6 marks)

An americium-241 atom, initially at rest, undergoes radioactive decay. The decay is shown with the mass of each atom in atomic mass units, u.



The kinetic energy of the alpha particle is $8.86 \times 10^{-13} \text{ J}$.

- (a) Calculate the kinetic energy of the Np-237 atom.

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- (b) The half-life of Am-241 is 432 years. A sample containing 6.4×10^{10} Am-241 atoms is to be left until 3.2×10^8 Am-241 atoms remain in the sample.

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How long will this take?

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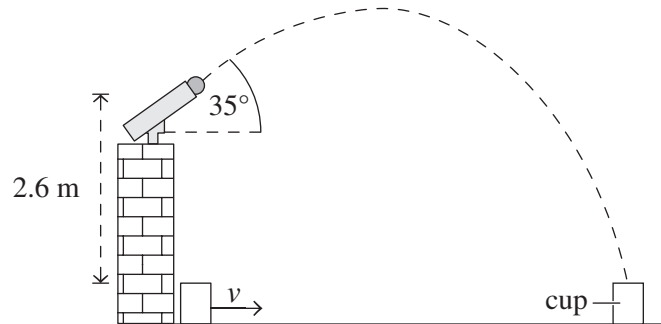
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Question 35 (7 marks)

A spring-loaded ball launcher is fixed at a 35° angle on a platform, as shown in the diagram. When a 46 g ball is launched, a target cup is released from the base of the platform and moves along a linear track directly away from the platform at a constant speed. The speed at which the target cup moves along the track can be set by the operator prior to launching the ball.

7



It is known that the spring in the launcher does 19.4 J of work on the ball.

Determine the required speed of the target cup and the distance the target cup must travel so that the ball lands in the cup.

[illegible]

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 ms^{-1}
Earth's gravitational acceleration, g	9.8 ms^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ ms}^{-1}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Wien's displacement constant, b	$2.898 \times 10^{-3} \text{ m K}$

FORMULAE SHEET

Motion, forces and gravity

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\Delta U = mg\Delta h$$

$$W = F_{\parallel}s = Fs \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$K = \frac{1}{2}mv^2$$

$$\sum \frac{1}{2}mv^2_{\text{before}} = \sum \frac{1}{2}mv^2_{\text{after}}$$

$$P = F_{\parallel}v = Fv \cos \theta$$

$$\Delta \vec{p} = \vec{F}_{\text{net}}\Delta t$$

$$\sum m\vec{v}_{\text{before}} = \sum m\vec{v}_{\text{after}}$$

$$\omega = \frac{\Delta \theta}{t}$$

$$a_c = \frac{v^2}{r}$$

$$\tau = r_{\perp}F = rF \sin \theta$$

$$F_c = \frac{mv^2}{r}$$

$$v = \frac{2\pi r}{T}$$

$$F = \frac{GMm}{r^2}$$

$$U = -\frac{GMm}{r}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

Waves and thermodynamics

$$v = f\lambda$$

$$f_{\text{beat}} = |f_2 - f_1|$$

$$f = \frac{1}{T}$$

$$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$$

$$d \sin \theta = m\lambda$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_x = \frac{c}{v_x}$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$I = I_{\text{max}} \cos^2 \theta$$

$$I_1 r_1^2 = I_2 r_2^2$$

$$Q = mc\Delta T$$

$$\frac{Q}{t} = \frac{kA\Delta T}{d}$$

FORMULAE SHEET (CONTINUED)**Electricity and magnetism**

$$E = \frac{V}{d}$$

$$\vec{F} = q\vec{E}$$

$$V = \frac{\Delta U}{q}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$W = qV$$

$$I = \frac{q}{t}$$

$$W = qEd$$

$$V = IR$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$P = VI$$

$$B = \frac{\mu_0 NI}{L}$$

$$F = qv_{\perp} B = qvB \sin \theta$$

$$\Phi = B_{\parallel} A = BA \cos \theta$$

$$F = I l_{\perp} B = I l B \sin \theta$$

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\tau = n I A_{\perp} B = n I A B \sin \theta$$

$$V_p I_p = V_s I_s$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$t = \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$K_{\max} = hf - \phi$$

$$l = l_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

$$\lambda_{\max} = \frac{b}{T}$$

$$p_v = \frac{m_0 v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$E = mc^2$$

$$N_t = N_0 e^{-\lambda t}$$

$$E = hf$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$



Trial Examination 2025

HSC Year 12 Physics

Solutions and Marking Guidelines

SECTION I**Question 1 D**

D is correct. The time of flight of a ball that has rolled off a horizontal bench is dependent only on the height of the bench. Decreasing the height of the bench would decrease the time of flight.

A and **B** are incorrect. As the ball is launched horizontally, the speed of the ball has no impact on the time of flight.

C is incorrect. Increasing the height of the bench would increase the time of flight.

Mod 5 Advanced Mechanics (PH12–12)

Bands 2–3

Question 2 A

Negatively charged particles experience a force opposite to the direction of the electric field. Given that the electric field lines are pointing right, the negatively charged particle will experience a force to the left.

Mod 6 Electromagnetism (PH12–13)

Bands 2–3

Question 3 C

An object in a stable orbit has a constant total mechanical energy. As the object in orbit moves closer to the body being orbited, its gravitational potential energy transforms into kinetic energy, and its kinetic energy turns into gravitational potential energy. In accordance with $K = \frac{1}{2}mv^2$, higher kinetic energy corresponds to a higher speed.

As position *Y* is the closest point to the star being orbited, the comet will have the greatest kinetic energy, and thus the greatest speed, at this position.

Mod 5 Advanced Mechanics (PH12–12)

Bands 2–3

Question 4 B

$$\begin{aligned}\lambda_{\max} &= \frac{b}{T} \\ &= \frac{2.898 \times 10^{-3}}{3500} \\ &= 8.28 \times 10^{-7} \text{ m}\end{aligned}$$

Mod 7 The Nature of Light (PH12–14)

Bands 2–3

Question 5 B

B is correct. Hadrons are a class of subatomic particles made of two or more quarks and are therefore subject to the strong interaction. Protons are made of three quarks (up, up, down); hence, they are hadrons.

A and **C** are incorrect. Leptons are elementary particles with half-integer spin and are not subject to the strong interaction; muons are a specific type of charged lepton.

D is incorrect. Photons are elementary particles that carry the electromagnetic force.

Mod 8 From the Universe to the Atom (PH12–15)

Bands 2–3

Question 6 B

The interaction between the two current-carrying wires is given by $\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$.

Therefore, $F \propto \frac{1}{r}$. Hence, increasing r to $2r$ gives $\frac{F}{2}$.

Mod 6 Electromagnetism (PH12–5, 12–13)

Bands 3–4

Question 7 B

The activity ratio will remain at 3, regardless of how many half-lives occur.

Mod 8 From the Universe to the Atom (PH12–15)

Bands 3–4

Question 8 B

In a DC motor, the commutator reverses the current direction in the rotor every half-turn. This maintains a unidirectional torque and, hence, ensures continuous rotational motion.

Mod 6 Electromagnetism (PH12–13)

Bands 3–4

Question 9 C

C is correct. According to Faraday's Law, $\varepsilon \propto N \frac{\Delta(B_{\parallel}A)}{\Delta t}$. Thus, halving the speed of rotation of the coil will result in an output emf that is half the previous output emf and will double the time it takes to complete one cycle. This matches the dashed line.

A, **B** and **D** are incorrect. While each of these modifications will result in an output emf that is half the previous output emf, they will not extend one cycle from 0.1 seconds to 0.2 seconds, as shown by the dashed line.

Mod 6 Electromagnetism (PH12–5, 12–13)

Bands 3–4

Question 10 D

According to $\lambda = \frac{h}{mv}$, the de Broglie wavelength is directly proportional to the reciprocal of speed;

that is, $\lambda \propto \frac{1}{v}$.

Mod 8 From the Universe to the Atom (PH12–5, 12–15)

Bands 4–5

Question 11 D

A nucleus that emits an electron and an antineutrino has undergone beta-minus decay. This process changes a neutron in the nucleus into a proton, increasing the atomic number by one. In this case, plutonium (atomic number 94) becomes americium (atomic number 95). During beta decay, mass stays constant. Therefore, the mass of the daughter nucleus is still 241 after the decay.

Mod 8 From the Universe to the Atom (PH12–15)

Bands 4–5

Question 12 A

A is correct. When the coils of an operating simple DC motor are suddenly stopped by an external force, the back emf decreases. As net emf = supply emf – back emf, the net emf increases, which results in an increase in current.

B is incorrect. Net emf would increase due to the decrease in back emf.

C is incorrect. Back emf would decrease, not increase.

D is incorrect. Back current is not a valid term.

Mod 6 Electromagnetism (PH12–13)

Bands 4–5

Question 13 A

$$\begin{aligned} v_{\max} &= \frac{2\pi r}{T} \\ &= \frac{2\pi \times 0.06}{\frac{1}{3.5}} \\ &= 1.3 \text{ m s}^{-1} \end{aligned}$$

Mod 5 Advanced Mechanics (PH12–12)

Bands 4–5

Question 14 **A**

The proton–proton chain is dominant in the cores of main sequence stars with low mass, which are found in region *R*. Region *S* contains main sequence stars with high mass; the dominant fusion process in the cores of these stars is the carbon–nitrogen–oxygen (CNO) cycle.

The largest stars on the diagram are the red supergiants, which are located in region *Q*. While regions *S* and *Q* have the same luminosity, the lower temperature of stars in region *Q* indicates a much larger size.

Mod 8 From the Universe to the Atom (PH12–15)

Bands 4–5

Question 15 **C**

$$\begin{aligned}
 E &= mc^2 \\
 &= 2 \times (9.109 \times 10^{-31}) \times (3.00 \times 10^8)^2 \\
 &= 1.6396 \times 10^{-13} \text{ J} \\
 \frac{1.6396 \times 10^{-13}}{1.602 \times 10^{-19}} &= 1.0 \times 10^6 \text{ eV} \\
 &= 1.0 \text{ MeV}
 \end{aligned}$$

As two gamma ray photons are produced in a positron-electron annihilation, each photon has 0.5 MeV of energy.

Mod 7 The Nature of Light (PH12–6, 12–14)

Bands 5–6

Question 16 **C**

C is correct. Induced emf is given by $\varepsilon = -N \frac{\Delta(BA \cos \theta)}{\Delta t}$. In this equation, θ refers to the angle between the magnetic field and the area vector, whereas in the question, θ refers to the angle between the magnetic field and the plane of the loop. Therefore, the minimum change in magnetic flux, and subsequently the minimum induced emf, will occur when the angle between the plane of the coil and the loop is 90° .

A is incorrect. The induced emf is at a maximum when the angle is 0° .

B is incorrect. The induced emf is 71% of the maximum when the angle is 45° .

D is incorrect. The induced emf varies throughout the coil's rotation.

Mod 6 Electromagnetism (PH12–6, 12–13)

Bands 5–6

Question 17 B

$$\frac{r_A^3}{T_A^2} = \frac{r_B^3}{T_B^2}$$

$$T_B = \sqrt{\frac{r_B^3 T_A^2}{r_A^3}}$$

$$= \sqrt{\frac{1^3 \times 1^2}{6^3}}$$

$$= 0.0680$$

$$\frac{1}{0.0680} = 14.7$$

Therefore, $T_A = 15T_B$.

Mod 5 Advanced Mechanics (PH12–6, 12–12)

Bands 5–6

Question 18 B

Finding the height of Mount Everest from the muon's perspective gives:

$$l = l_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

$$= 8.8 \sqrt{1 - 0.97^2}$$

$$= 2.13 \dots \text{ km}$$

From the muon's perspective, the event lasts for 2.2×10^{-6} s.

Therefore, from the muon's perspective:

$$\text{distance} = vt$$

$$= 0.97 \times (3.00 \times 10^8) \times (2.2 \times 10^{-6})$$

$$= 640.2 \text{ m}$$

$$= 0.6402 \text{ km}$$

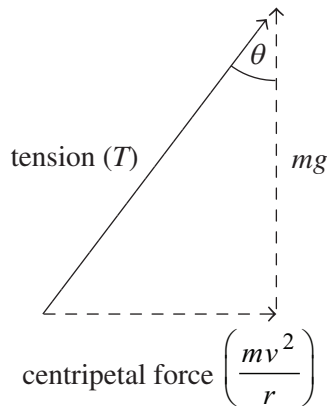
$$2.13 \dots - 0.6402 = 1.5 \text{ km}$$

Mod 7 The Nature of Light (PH12–6, 12–14)

Bands 5–6

Question 19 A

A is correct and **D** is incorrect.



$$\theta = \tan^{-1}\left(\frac{mv^2}{rmg}\right) = \tan^{-1}\left(\frac{v^2}{rg}\right)$$

B and **C** are incorrect. To calculate θ using the sine function, $\sin^{-1}\left(\frac{mv^2}{rT}\right)$ would be the correct expression.

Mod 5 Advanced Mechanics (PH12–5, 12–12)

Bands 5–6

Question 20 D

D is correct. The energy difference between $n = 4$ and $n = 2$ (2.55 eV) is smaller than the energy difference between $n = 2$ and $n = 1$ (10.2 eV). Hence, knowing that $E = \frac{hc}{\lambda}$, the wavelength of a photon produced from an electron transitioning from $n = 4$ to $n = 2$ will be longer than the wavelength of a photon produced by an electron transitioning from $n = 2$ to $n = 1$.

A is incorrect. An electron requires 2.86 eV of energy to transition from $n = 2$ to $n = 5$. It will only produce a photon of 2.86 eV if it transitions from $n = 5$ to $n = 2$.

B is incorrect. The Balmer series includes spectral emission lines that result from electron transitions from higher levels down to $n = 2$, not $n = 1$.

C is incorrect. The energy required to ionise a hydrogen atom when its electron is in the ground state (that is, $n = 1$) is four times greater than the energy required to ionise the atom when its electron is excited to the $n = 2$ state (13.6 eV is four times greater than 3.4 eV). However, this statement refers to transitioning an electron from $n = 1$ to $n = 2$, which requires 10.2 eV of energy. Therefore, ionising a hydrogen atom when its electron is in the ground state requires 1.33 times more energy than moving an electron from $n = 1$ to $n = 2$.

Mod 8 From the Universe to the Atom (PH12–6, 12–12)

Bands 5–6

SECTION II

Question 21

- (a) The magnitude of the forces acting on sides AB and CD in the position shown are the same. The direction of the force acting on side AB is into the page, whereas the direction of the force acting on side CD is out of the page.

Mod 6 Electromagnetism (PH12–5, 12–13)

Bands 3–4

- Compares the magnitude and direction of the forces acting on sides AB and CD 2
- Provides some relevant information 1

- (b) In accordance with $\tau = nIA \perp B$, doubling the number of loops (n) would double the maximum torque (τ) produced by the motor.

Mod 6 Electromagnetism (PH12–13)

Bands 3–4

- Outlines the impact on the maximum torque.
AND
- Supports the answer with a relevant equation 2
- Provides some relevant information 1

Question 22

(a)
$$F_c = \frac{mv^2}{r}$$

$$= \frac{m \left(\frac{2\pi r}{T} \right)^2}{r}$$

$$= \frac{0.031 \times \left(\frac{2\pi \times 0.35}{1} \right)^2}{0.35}$$

$$= 0.43 \text{ N}$$

Mod 5 Advanced Mechanics (PH12–12)

Bands 3–4

- Calculates the centripetal force 3
- Calculates the centripetal force with ONE error or omission 2
- Provides some relevant working 1

- (b) Regardless of where the object is placed on the disc, its angular speed will remain constant. Hence, the object's angular speed will remain at $2\pi \text{ rad s}^{-1}$ if the object was moved to a point closer to the centre of the disc. This is in accordance with $\omega = \frac{\Delta\theta}{t}$.

Mod 5 Advanced Mechanics (PH12–12)

Bands 3–4

- States that the angular speed is constant at all points on the disc.
AND
- Supports the answer with a relevant equation 2
- Provides some relevant information 1

Question 23

(a) $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ and $V_p I_p = V_s I_s$

Therefore:

$$I_p = \frac{N_s I_s}{N_p}$$

$$= \frac{12 \times 5}{6}$$

$$= 10 \text{ A}$$

Mod 6 Electromagnetism (PH12–13)

Bands 3–4

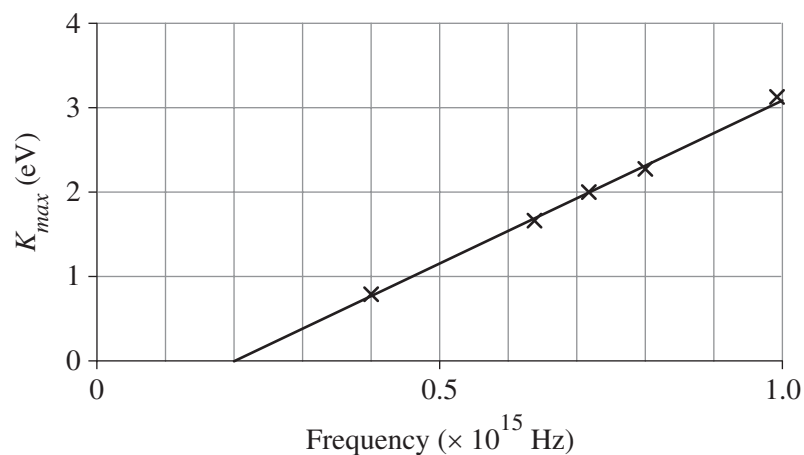
- Shows that the current in the primary coil is 10 A 3
- Provides some correct steps. 2
- Provides some relevant information 1

- (b) Transformers operate on the principle of electromagnetic induction, which relies on a changing magnetic field. AC produces a continuous change in current direction, creating a fluctuating magnetic field that is essential for inducing voltage in the secondary coil. On the other hand, DC provides a constant current that does not create this changing magnetic field, rendering transformers ineffective.

Mod 6 Electromagnetism (PH12–13)

Bands 4–5

- Explains why it is important to use AC rather than DC when operating transformers 3
- Shows some understanding of electromagnetic induction AND the difference between AC and DC 2
- Provides some relevant information 1

Question 24

threshold frequency = x -intercept = 0.2×10^{15} Hz

Finding the frequency of incident light gives:

$$\begin{aligned}
 f &= \frac{c}{\lambda} \\
 &= \frac{3.00 \times 10^8}{3 \times 10^{-6}} \\
 &= 0.1 \times 10^{15} \text{ Hz}
 \end{aligned}$$

As 0.1×10^{15} Hz is less than the threshold frequency of 0.2×10^{15} Hz, photoelectrons would not be emitted if the metal was exposed to this light.

Note: Accept answers within the range 0.16–0.24.

Mod 7 The Nature of Light (PH12–6, 12–14)

Bands 4–5

- Plots all data points.
AND
- Draws a line of best fit.
AND
- Determines that photoelectrons are not emitted based on a relevant calculation 4

- Plots all data points.
AND
- Provides some correct steps in the calculation 3

- Plots all data points with ONE error or omission.
AND
- Provides ONE correct step in the calculation 2

- Provides some relevant information 1

Question 25

The quantum mechanical model describes atomic orbitals, which are regions of space where electrons are likely to be found. Through the development of his wave equation, Schrödinger contributed to this model by establishing the concept of the wave function, which provides a mathematical description of an electron's state in an atom. It was then demonstrated that squaring the absolute value of the wave function gives the probability density, indicating the likelihood of finding an electron in a specific region of space within the atom.

Mod 8 From the Universe to the Atom (PH12–15)

Bands 4–5

- Outlines Schrödinger's contribution to the quantum mechanical model of the atom 3

- Outlines some features of the quantum mechanical model of the atom 2

- Provides some relevant information 1

Question 26

- (a) The atomic clocks on the eastward flight (with Earth's rotation) were moving faster relative to the atomic clocks on the ground. The atomic clocks on the westward flight (against Earth's rotation) were moving slower relative to the atomic clocks on the ground. The theory of special relativity predicts that the atomic clocks on the eastward flight would show slightly less elapsed time compared to the atomic clocks on the ground, while the atomic clocks on the westward flight would show slightly greater elapsed time. The results of the Hafele-Keating experiment matched the predictions, thus validating time dilation as predicted by special relativity.

Mod 7 The Nature of Light (PH12–14)

Bands 4–5

- Explains how the Hafele-Keating experiment validated time dilation as predicted by special relativity 3

- Outlines the relative speed and time predictions for ONE flight.
OR
- Outlines the time predictions for both flights 2

- Provides some relevant information 1

- (b) According to classical physics, momentum (p) is calculated by multiplying an object's mass (m) and its velocity (v). Therefore, the velocity of an electron with a momentum of $8.7 \times 10^{-22} \text{ kg m s}^{-1}$ would be:

$$\begin{aligned} v &= \frac{p}{m} \\ &= \frac{8.7 \times 10^{-22}}{9.109 \times 10^{-31}} \\ &= 9.6 \times 10^8 \text{ m s}^{-1} \end{aligned}$$

This speed is not possible as it exceeds the speed of light. Hence, classical physics cannot account for this momentum. Relativistic effects need to be considered to account for the electron's momentum.

Mod 7 The Nature of Light (PH12–6, 12–14)

Bands 4–5

- Explains why classical physics cannot account for the electron's momentum.
AND
- Supports the answer with a relevant calculation 3

- Shows some understanding of the limitations of classical physics 2

- Provides some relevant information 1

Question 27

(a)
$$\begin{aligned} U &= -\frac{GMm}{r} \\ &= -\frac{(6.67 \times 10^{-11}) \times (9.1 \times 10^{20}) \times 900}{476 \times 10^3} \\ &= -1.1 \times 10^8 \text{ J} \end{aligned}$$

Mod 5 Advanced Mechanics (PH12–12)

Bands 3–4

- Shows the gravitational potential energy would be $-1.1 \times 10^8 \text{ J}$ 2

- Provides some relevant information 1

(b) **Method 1:**

$$\frac{1}{2}mv^2 = 1.1 \times 10^8$$

$$v = \sqrt{\frac{2 \times (1.1 \times 10^8)}{900}}$$

$$= 494 \text{ m s}^{-1}$$

Method 2:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$$

$$= \sqrt{\frac{2 \times (6.67 \times 10^{-11}) \times (9.1 \times 10^{20})}{476 \times 10^3}}$$

$$= 505 \text{ m s}^{-1}$$

Mod 5 Advanced Mechanics (PH12–12)

Bands 4–5

- Calculates the escape velocity 3

- Calculates the escape velocity with ONE error or omission 2

- Provides some relevant working 1

Question 28

- (a) Stars produce light across a broad spectrum, including visible light. As light travels outward from this star, it passes through the star's cooler outer layers, where hydrogen atoms are present. Hydrogen atoms absorb specific wavelengths of light that correspond to the energy differences between their electron energy levels. When photons of specific wavelengths are absorbed by these hydrogen atoms, dark lines are created in the star's otherwise continuous spectrum. The result is a spectrum that shows distinct hydrogen absorption lines.

Mod 8 From the Universe to the Atom (PH12–7, 12–15)

Bands 4–5

- Accounts for the production of the spectrum 4

- Accounts for the production of the spectrum with ONE error or omission 3

- Provides ONE reason for the spectrum 2

- Provides some relevant information 1

- (b) The star's translational velocity can be determined using the shifts in its hydrogen absorption lines. The wavelengths of the absorption lines at rest are known. Due to the relative motion of the star, the wavelengths of the absorption lines can shift as the star moves. If the star is moving towards the observer, the absorption lines will shift towards the blue end of the spectrum. If the star is moving away from the observer, the absorption lines will shift towards the red end. The velocity of the star can then be calculated from these shifts in wavelengths. The greater the shift, the greater the velocity.

Mod 7 The Nature of Light (PH12–14)

Bands 4–5

- Explains how the translational velocity of the star can be determined from its spectrum 3
- Shows some understanding of translational velocity and the shifting of absorption lines 2
- Provides some relevant information 1

Question 29

$$\begin{aligned}
 F_c &= F_B \\
 \frac{mv_{\perp}^2}{r} &= qv_{\perp}B \\
 \frac{q}{m} &= \frac{v \sin \theta}{rB} \\
 &= \frac{(2.50 \times 10^6) \times \sin(62.0)}{0.286 \times (160 \times 10^{-3})} \\
 &= 4.824 \times 10^7 \text{ C kg}^{-1}
 \end{aligned}$$

Mod 6 Electromagnetism (PH12–6, 12–13)

Bands 5–6

- Calculates the charge-to-mass ratio of the particle 4
- Calculates the charge-to-mass ratio with ONE error or omission 3
- Provides some correct steps 2
- Provides some relevant working 1

Question 30

Polarisation plays a crucial role in our understanding and modelling of light. When light passes through certain materials (such as polarising filters), it becomes polarised, which means that it oscillates in specific directions rather than in all directions. This directional behaviour provides strong evidence that light is a transverse wave, not a longitudinal wave as Huygens had predicted.

Maxwell's electromagnetic theory of light describes light as oscillating electric and magnetic fields. The theory predicts that these fields oscillate perpendicular to the direction of propagation, which aligns with the observed behaviour of polarised light and reinforces the transverse wave model. Only transverse light waves with an electric field parallel to the polariser can pass through, while perpendicular components will be blocked.

Mod 7 The Nature of Light (PH12–7, 12–14)**Bands 5–6**

- Provides a comprehensive evaluation of the significance of polarisation in developing a transverse wave model of light 6

- Provides an evaluation of the significance of polarisation in developing a transverse wave model of light 5

- Relates polarisation to features of the transverse wave model of light 4

- Outlines polarisation.
AND
- Outlines the transverse wave model of light 3

- Outlines polarisation.
OR
- Outlines the transverse wave model of light 2

- Provides some relevant information 1

Question 31

Given that $d\sin\theta = m\lambda$ and $f \propto \frac{1}{\lambda}$, increasing the frequency of the incident light will decrease the angle, θ . This will also decrease the distance between the bright regions on the screen.

To maintain the distance between the bright regions on the screen, the distance between the slits and the screen can be increased. This would counteract the decrease of θ caused by increasing the frequency of the incident light.

Another modification involves decreasing the distance, d , between the slits. This increases θ and thus compensates for the increase in the frequency of the incident light.

Mod 7 The Nature of Light (PH12–6, 12–14)		Bands 5–6
• Explains TWO modifications	4	
• Explains ONE modification. AND		
• Outlines a second modification	3	
• Explains ONE modification. OR		
• Identifies TWO modifications	2	
• Provides some relevant information	1	

Question 32

$$I = \frac{V}{R}$$

$$= \frac{50}{5}$$

= 10 A (in each circuit)

$$F = l \times \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$= 0.50 \times \frac{4\pi \times 10^{-7}}{2\pi} \frac{10 \times 10}{0.08}$$

= 1.25×10^{-4} N attractive (currents are in same direction)

$$a = \frac{F_{\text{net}}}{m}$$

$$= \frac{1.25 \times 10^{-4}}{0.65}$$

$$= 1.92 \times 10^{-4} \text{ m s}^{-2}$$

initial acceleration of cart A = $1.92 \times 10^{-4} \text{ m s}^{-2}$ to the right

initial acceleration of cart B = $1.92 \times 10^{-4} \text{ m s}^{-2}$ to the left

Mod 6 Electromagnetism (PH12–6, 12–13)

Bands 5–6

- Determines the magnitude AND direction of the initial acceleration of each cart5
- Determines the magnitude of the initial acceleration of each cart4
- Calculates the magnitude of the force acting on each rod.3
- Calculates the current in each circuit.2
- Provides some relevant information1

Question 33

The nuclear model of the atom posits that atoms have a small, dense nucleus containing protons and neutrons. This model is supported by experimental evidence from the Geiger–Marsden experiment and Chadwick’s discovery of the neutron.

In 1909, under the direction of Rutherford, Geiger and Marsden conducted the gold foil experiment. In this experiment, alpha particles were directed at a thin sheet of gold foil. According to the Thomson atomic model, which was accepted at the time, all the alpha particles were expected to pass straight through the foil. Instead, it was observed that a small percentage of the particles were deflected at large angles, with some even rebounding. This result led Rutherford to propose that atoms have a dense, positively charged core where most of its mass is concentrated, while the rest of the atom is mostly empty space. The large-angle deflections occurred when alpha particles passed close to or collided with this core in the gold atoms. This experiment provided the first direct evidence for the existence of an atomic nucleus.

In 1920, Rutherford hypothesised the existence of a neutral particle to account for the mass discrepancy in atomic nuclei. In 1932, Chadwick discovered this particle. Chadwick bombarded beryllium with alpha particles, leading to the emission of uncharged radiation. This uncharged radiation was then directed towards a block of paraffin wax, ejecting protons. Using conservation laws (specifically energy and momentum), he demonstrated that the radiation consisted of neutral particles with a mass similar to that of protons; thus discovering the neutron. Chadwick’s discovery of the neutron provided further insight into the composition of the nucleus.

Together, these experiments firmly established the nuclear model, which states that the nucleus consists of protons and neutrons, with electrons orbiting the nucleus in the surrounding space.

Mod 8 From the Universe to the Atom (PH12–7, 12–15)

Bands 5–6

- Comprehensively analyses how the Geiger–Marsden experiment and Chadwick’s discovery of the neutron provide evidence for the nuclear model of the atom 8

- Analyses how the Geiger–Marsden experiment and Chadwick’s discovery of the neutron provide evidence for the nuclear model of the atom 7

- Analyses how the Geiger–Marsden experiment provides evidence for the nuclear model of the atom AND describes Chadwick’s discovery of the neutron.
OR
- Analyses how Chadwick’s discovery of the neutron provides evidence for the nuclear model of the atom AND describes the Geiger–Marsden experiment. 6

- Relates the features of the nuclear model of the atom to both experiments 5

- Provides some links between the nuclear model of the atom and both experiments 4

- Outlines the nuclear model of the atom.
AND
- Outlines ONE of the experiments 3

- Outlines the nuclear model of the atom.
OR
- Outlines ONE of the experiments 2

- Provides some relevant information 1

Question 34

(a) $\Delta m = 241.057 - (237.048 + 4.00260)$
 $= 0.0064 \text{ u}$
 $E = \Delta mc^2$
 $= (0.0064 \times 1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$
 $= 9.56 \dots \times 10^{-13} \text{ J}$
 $K_{\text{Np-237}} = (9.56 \dots \times 10^{-13}) - (8.86 \times 10^{-13})$
 $= 7 \times 10^{-14} \text{ J}$

Mod 8 From the Universe to the Atom (PH12–15)

Bands 4–5

- Calculates the kinetic energy of the Np-237 atom 3

- Calculates the total kinetic energy of the decay 2

- Provides some relevant working 1

(b) $\lambda = \frac{\ln(2)}{t_1}$
 $\frac{2}{2}$
 $= \frac{\ln(2)}{432}$
 $= 0.0016 \dots \text{ years}^{-1}$
 $N_t = N_0 e^{-\lambda t}$
 $t = \frac{\ln\left(\frac{3.2 \times 10^8}{6.4 \times 10^{10}}\right)}{-0.0016 \dots}$
 $= 3300 \text{ years}$

Mod 8 From the Universe to the Atom (PH12–6, 12–15)

Bands 5–6

- Calculates the time it will take for 3.2×10^8 Am-241 atoms to remain 3

- Calculates the decay constant.
AND
- Attempts to calculate the time using the relevant formula 2

- Provides some relevant working 1

Question 35

Finding the launch velocity of the ball gives:

$$\begin{aligned} v &= \sqrt{\frac{2W}{m}} \\ &= \sqrt{\frac{2 \times 19.4}{0.046}} \\ &= 29.04 \dots \text{ m s}^{-1} \end{aligned}$$

The required speed of the target cup is equal to the horizontal component of the ball's launch velocity.

Therefore, the required speed is:

$$u_x = 29.04 \dots \cos(35) = 23.8 \text{ m s}^{-1}$$

Finding the time it takes to reach the top of the ball's trajectory gives:

$$\begin{aligned} t_{\text{up}} &= \frac{v_y - u_y}{a_y} \\ &= \frac{0 - (29.04 \dots \sin(35))}{-9.8} \\ &= 1.69 \dots \text{ s} \end{aligned}$$

Finding the vertical displacement from the ball's initial position to the top of its trajectory gives:

$$\begin{aligned} s_{\text{up}} &= u_y t_{\text{up}} + \frac{1}{2} a_y t_{\text{up}}^2 \\ &= (29.04 \dots \sin(35)) \times 1.69 \dots + \frac{1}{2} \times -9.8 \times 1.69 \dots^2 \\ &= 14.15 \dots \text{ m} \end{aligned}$$

Finding the vertical displacement from the top of the ball's trajectory to the target cup gives:

$$\begin{aligned} s_{\text{down}} &= 14.15 \dots + 2.6 \\ &= 16.75 \dots \text{ m} \end{aligned}$$

Finding the time from the top of the ball's trajectory to the target cup gives:

$$\begin{aligned} t_{\text{down}} &= \sqrt{\frac{2s_{\text{down}}}{a_y}} \\ &= \sqrt{\frac{2 \times 16.75 \dots}{9.8}} \\ &= 1.84 \dots \text{ s} \end{aligned}$$

Finding the total time of flight gives:

$$\begin{aligned} t &= t_{\text{up}} + t_{\text{down}} \\ &= 1.69 \dots + 1.84 \dots \\ &= 3.54 \dots \text{ s} \end{aligned}$$

Therefore, the distance the target cup must travel is:

$$\begin{aligned} s_x &= u_x t \\ &= 23.8 \times 3.54 \dots \\ &= 84 \text{ m} \end{aligned}$$

(continues on next page)

(continued)

Mod 5 Advanced Mechanics (PH12–6, 12–12)	Bands 5–6
Calculates the required speed of and distance travelled by the target cup	7
Calculates the required speed of and distance travelled by the target cup with ONE error or omission	6
- Calculates the required speed of the target cup AND the total time of flight. OR - Calculates the required distance travelled by the target cup	5
- Determines the required speed of the target cup. AND - Provides some further correct steps.	4
- Determines the required speed of the target cup. OR - Calculates the launch velocity of the ball and provides a further correct step	3
Calculates the launch velocity of the ball	2
Provides some relevant working	1